Differentiable Sampling of Loop-Quantum-Gravity Horizon Microstate Ensembles: A Continuous Relaxation and Hamiltonian Monte Carlo Framework

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Abstract

This report formulates and evaluates a differentiable probabilistic surrogate for the fixed-puncture area ensemble of a loop-quantum-gravity (LQG) black-hole horizon. The exact kinematic labels are half-integer spins, whereas Hamiltonian Monte Carlo (HMC) requires a smooth target on a connected continuous state space. Each spin is therefore represented by x_i = 1/2 + softplus(z_i), with an area-shell penalty, a periodic half-integer comb, and a factorized degeneracy pressure. A change-of-variables analysis shows that the negative log Softplus Jacobian is mandatory: without it the latent density is improper as any coordinate approaches negative infinity. The comb has finite, smooth barriers and local curvature 8 pi^2 kappa_Q; it concentrates mass near valid quantum labels but neither produces exact discreteness nor guarantees rapid movement among spin sectors. A custom PyTorch implementation uses exact autograd gradients, reversible leapfrog trajectories, Metropolis correction, dual-averaging step-size adaptation, and diagonal mass adaptation. In the validated N = 64 baseline, acceptance was .925 with zero divergences. Mean area was 523.405 against a target of 512, area effective sample size (ESS) was 36.22 of 80, and lattice-distance ESS was 18.34 of 80. These findings establish numerical feasibility but not black-hole entropy: the local product of representation dimensions is not the invariant intertwiner multiplicity, the normalizing constant is unknown, and a single short chain cannot certify inter-well mixing. A confirmatory program based on replica tempering, low-rank geometry, exact small-N enumeration, and thermodynamic integration is specified.

Keywords: loop quantum gravity, black-hole entropy, continuous relaxation, Hamiltonian Monte Carlo, PyTorch, microstate counting

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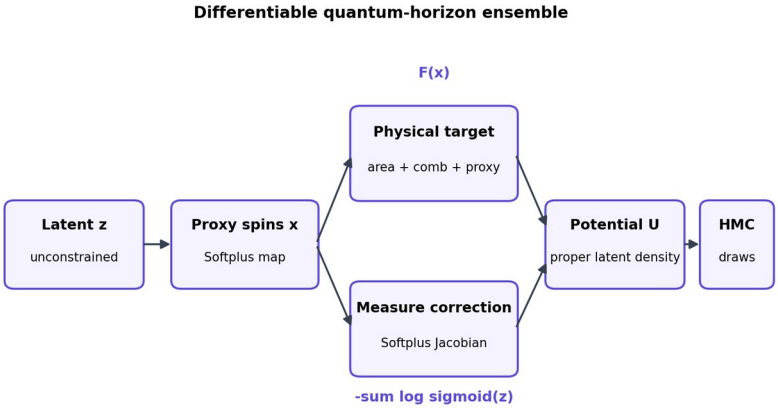
Research Problem and Contribution

Loop quantum gravity replaces classical horizon area by a discrete spectrum generated by spin-network edges that puncture an isolated horizon. The microscopic counting problem is consequently combinatorial: one must identify spin assignments compatible with a macroscopic area and then count admissible boundary or invariant states. Foundational calculations recover an area-proportional entropy with model-dependent logarithmic corrections (Ashtekar et al., 1998; Engle, Noui, & Perez, 2010). Exact enumeration becomes difficult when the number of punctures and admissible spins are large. Standard single-coordinate Metropolis updates are especially vulnerable to area constraints and coordinated changes among spin sectors.

The present project asks whether a smooth proxy can turn the discrete configuration problem into a gradient-based statistical experiment while preserving an explicit route back to the quantum lattice. Its contribution is not a new derivation of the Bekenstein-Hawking law. It is a computational architecture: a proper continuous density, an auditable HMC sampler, scalable curvature diagnostics, and a falsification protocol separating surrogate behavior from exact LQG state counting.

Figure 1

Computational and Statistical Architecture



LQG Horizon Kinematics and the Exact Counting Target

For a fixed number N of punctures, each puncture carries an irreducible SU(2) representation with j_i in {1/2, 1, 3/2, ...}. In Planck units the standard area contribution is proportional to sqrt[j_i(j_i+1)], so the total eigenvalue is

$$A(\mathbf{j}) = 8\pi\gamma\ell_p^2 \sum_{i=1}^{N} \sqrt{j_i(j_i+1)} \qquad (1)$$

where gamma is the Barbero-Immirzi parameter. The macroscopic ensemble is a thin area window or an equivalent controlled statistical ensemble, not a single unconstrained product measure. Moreover, the physical boundary multiplicity depends on horizon constraints. In the SU(2)-invariant formulation it is related to Chern-Simons Hilbert spaces and, in an appropriate regime, invariant intertwiners (Engle et al., 2010; Engle et al., 2011).

Table 1

Exact and Relaxed Components of the Model

Component	Mathematical role	Interpretive status
Discrete spins j_i	Half-integer horizon labels	Exact LQG kinematics
Proxy spins x_i	Smooth positive relaxation	Computational surrogate
Area penalty	Soft target-area shell	Not a hard microcanonical constraint
Periodic comb	Concentrates near half-integers	Finite-barrier regularizer
Product (2x_i+1)^eta	Local degeneracy pressure	Not SU(2) intertwiner dimension
Softplus Jacobian	Defines correct latent measure	Required for propriety

Note. The relaxed sampler is a proposal and approximation architecture. Only the first row is an exact discrete LQG variable.

Discrete Ensemble, Intertwiners, and Entropy

Let J_N denote the half-integer lattice and let W_delta(A_star) be an area window. A schematic exact microcanonical count is

$$\Omega_N(A^*, \delta) = \sum_{\{\mathbf{j} \in \mathcal{J}_N\}} \mathbb{1}\{|A(\mathbf{j}) - A^*| \leq \delta\} \; d_phys(\mathbf{j}) \qquad (2)$$

$$d_phys(\mathbf{j}) = \dim \; Inv_SU(2)(\otimes_{i=1}^{N} V_\{j_i\}) \qquad (3)$$

with modifications when finite Chern-Simons level, projection constraints, distinguishability conventions, or distorted horizons are retained. Entropy is S = log Omega only after the ensemble and multiplicity are fully defined. The factor product_i(2j_i+1) counts dimensions before gauge-invariant coupling and generally overstates the invariant state space. Domagala and Lewandowski (2004) and Meissner (2004) also show that higher spins cannot be dismissed without checking the exact counting equation.

This distinction supplies the central statistical boundary of the project. The relaxed target may discover high-probability spin sectors, estimate relative basin weights, or construct proposals for a discrete correction. It cannot, by itself, identify Omega_N. The correct downstream estimand is therefore a calibrated discrete distribution or normalizing constant, not the posterior expectation of the proxy degeneracy term.

Relation to String-Theory Ensembles

String-theory microstate counts are typically organized by conserved charges and, in controlled cases, BPS degeneracies (Strominger & Vafa, 1996). Puncture spins are not interchangeable with those charge labels. A cross-theory comparison requires separately specified ensembles and matched macroscopic observables; no such identification is assumed here.

Continuous Relaxation and Change of Variables

The unconstrained state is z in R^N. Coordinatewise transformation gives

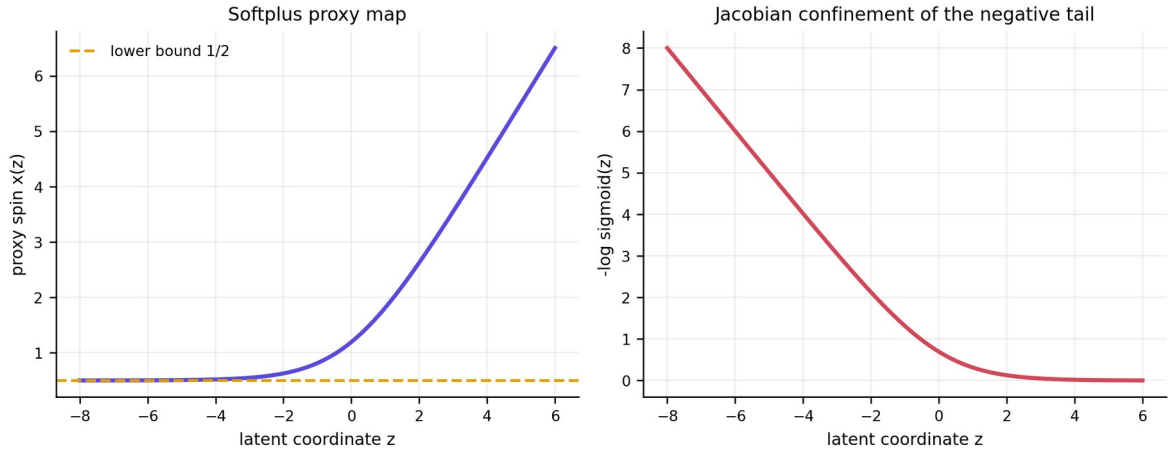
$$x_i = ½ + \mathrm{softplus}(z_i) = ½ + \log(1 + e^{z_i}) \qquad (4)$$

$$dx_i/dz_i = \sigma(z_i) \qquad (5)$$

The map is smooth, strictly increasing, and bijective from R to (1/2, infinity). It enforces the physical lower bound without projection or reflection. Unlike a Concrete or Gumbel-Softmax relaxation, it does not encode a categorical distribution; it creates a continuous proxy coordinate whose lattice preference must be supplied separately (Maddison et al., 2017; Zhang et al., 2012).

Figure 2

Softplus Proxy and the Required Jacobian Term



Note. The Jacobian is part of the probability model, not a numerical regularizer that may be omitted after tuning.

Potential Energy and Propriety of the Latent Density

Write C = 8 pi gamma l_p^2 and a(x)=sqrt[x(x+1)]. The physical proxy energy is

$$F(x) = (\kappa_A/2)[C\sum_i a(x_i) - A^*]^2 + \kappa_Q\sum_i \sin^2(2\pi x_i) - \eta\sum_i \log(2x_i+1) \qquad (6)$$

If exp[-F(x)] is a density with respect to dx, the pullback density with respect to dz has potential

$$U(z) = F(x(z)) - \sum_i \log\, \sigma(z_i) \qquad (7)$$

Proposition 1: Tail Coercivity

For positive kappa_A and finite nonnegative kappa_Q and eta, exp[-U(z)] is integrable. If z_i tends to negative infinity with other coordinates fixed, x_i-1/2 is asymptotic to exp(z_i), the physical energy approaches a finite limit, and -log sigmoid(z_i) is asymptotic to -z_i. Hence the latent density decays as exp(z_i). If any subset of coordinates tends to positive infinity, a(x_i) is asymptotically x_i+1/2 and the squared area residual grows quadratically in their sum. This dominates the bounded comb and logarithmic degeneracy reward. The same bounds extend to joint tails, establishing coercivity and integrability.

Omitting the Jacobian invalidates the first tail argument: U approaches a finite constant along z_i to negative infinity, leaving infinite Lebesgue mass. The corrected model therefore has a mathematically defined target for HMC, whereas the uncorrected version does not.

Periodic Comb Geometry and Quantum-Lattice Concentration

Define q(x)=sin^2(2 pi x). Its zero set is exactly the half-integer lattice because sin(2 pi x)=0 if and only if 2x is an integer. For h in (1/2)Z and delta near zero, Taylor expansion gives

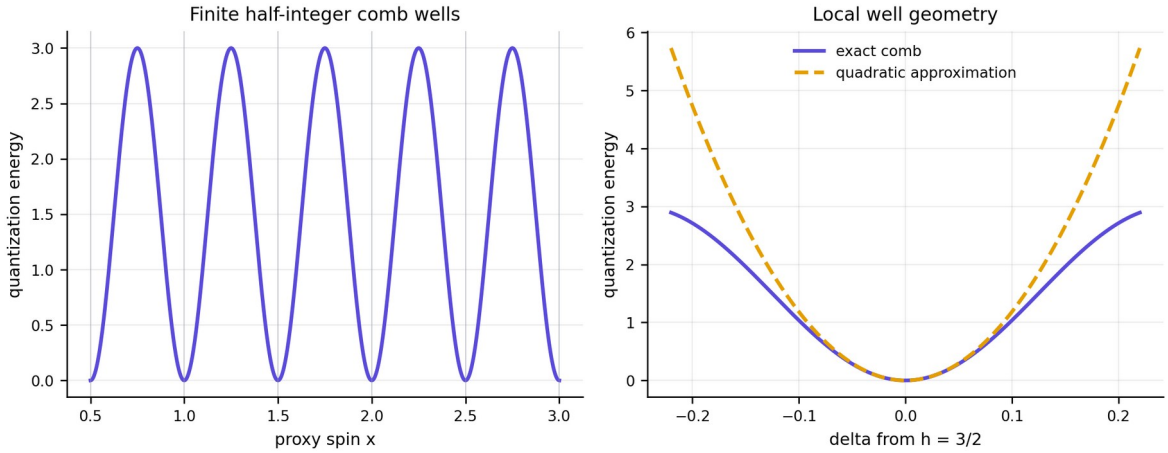
$$q(h+\delta) = 4\pi^2\delta^2 - (16/3)\pi^4\delta^4 + O(\delta^6) \qquad (8)$$

$$q''(h) = 8\pi^2 \qquad (9)$$

Thus the weighted local variance under a harmonic approximation is [8 pi^2 kappa_Q]^{-1}. At the baseline kappa_Q=3, the predicted local root-mean-square displacement is 0.06497. The barriers are finite: the single-coordinate maximum is kappa_Q, reached midway between adjacent lattice wells.

Figure 3

Periodic Comb and Local Harmonic Approximation



Nonabsorption Result

Every off-lattice point has finite potential and a smooth gradient. Under exact Hamiltonian flow with Gaussian momentum refreshment, the probability of drawing kinetic energy above any fixed finite barrier is positive. Consequently an unphysical region is not absorbing. This result rules out permanent trapping as a matter of support, but it does not imply practically useful mixing times.

Gradient Field, Area Shell, and Degeneracy Pressure

Let r=C sum_i a(x_i)-A_star and v_i=a'(x_i)=(2x_i+1)/[2 sqrt{x_i(x_i+1)}]. The physical gradient is

$$\partial F/\partial x_i = \kappa_A\ C\ r\ v_i + 2\pi\kappa_Q \sin(4\pi x_i) - 2\eta/(2x_i+1) \qquad (10)$$

$$\partial U/\partial z_i = \sigma(z_i)\ \partial F/\partial x_i + \sigma(z_i) - 1 \qquad (11)$$

The first force is collective: one coordinate changes the residual experienced by every other coordinate. The comb is oscillatory and local, while the proxy degeneracy term favors larger spins. The Jacobian adds a negative-tail restoring force. Their competition shifts finite-kappa_Q modes away from exact half-integers by order 1/kappa_Q; exact lattice recovery is therefore not obtained by merely rounding the posterior mean.

The quadratic area term defines a soft shell. Conditional on local linearization, shell width scales roughly as kappa_A^{-1/2}, but the effective width also depends on N, the collective gradient, and tangential spin multiplicity. A hard microcanonical window is a different target. Constrained HMC, auxiliary Lagrange multipliers, or a calibrated sequence kappa_A to infinity would be required to approximate it systematically.

Statistical Estimand

The immediate estimands are normalized relaxed expectations, area-shell error, lattice concentration, and transition efficiency. Physical entropy is deferred until basin normalization and exact invariant multiplicity are supplied. This ordering prevents a computationally convenient regularizer from being mistaken for a microscopic state count.

Hamiltonian Monte Carlo and Reversible Integration

Introduce momentum p distributed as N(0,M), where M is a positive-definite mass matrix. The Hamiltonian is

$$H(z,p) = U(z) + \tfrac{1}{2}\, p^{T}M^{-1}p \qquad (12)$$

A leapfrog step of size epsilon performs the symmetric updates

$$p_{n+1/2} = p_n - (\varepsilon/2)\nabla U(z_n) \qquad (13)$$

$$z_{n+1} = z_n + \varepsilon M^{-1} p_{n+1/2} \qquad (14)$$

$$p_{n+1} = p_{n+1/2} - (\varepsilon/2)\nabla U(z_{n+1}) \qquad (15)$$

Leapfrog is reversible and volume preserving. Numerical energy error is removed by accepting the proposal with

$$\alpha = \min\{1, \exp[-H(z',p')+H(z,p)]\} \qquad (16)$$

which preserves the target density despite discretization (Duane et al., 1987). The repository evaluates grad U with torch.autograd.grad and checks it against Equation 11. Momentum negation makes the proposal map an involution, integration length is randomized between 5 and 12 steps, and step jitter reduces resonance. Dual averaging and diagonal mass estimation occur only during warmup; adaptation is frozen before retained draws, preserving a homogeneous sampling kernel (Hoffman & Gelman, 2014; Roberts & Rosenthal, 2007).

Gradient clipping is deliberately excluded because it changes the Hamiltonian and invalidates the Metropolis correction unless the clipped field is declared to be a different potential. Numerical stability is instead handled through float64 arithmetic, finite-value checks, step-size control, and divergence diagnostics.

Statistical Geometry of the Relaxed Surface

In proxy coordinates the Hessian decomposes as a diagonal matrix plus one collective rank-one term:

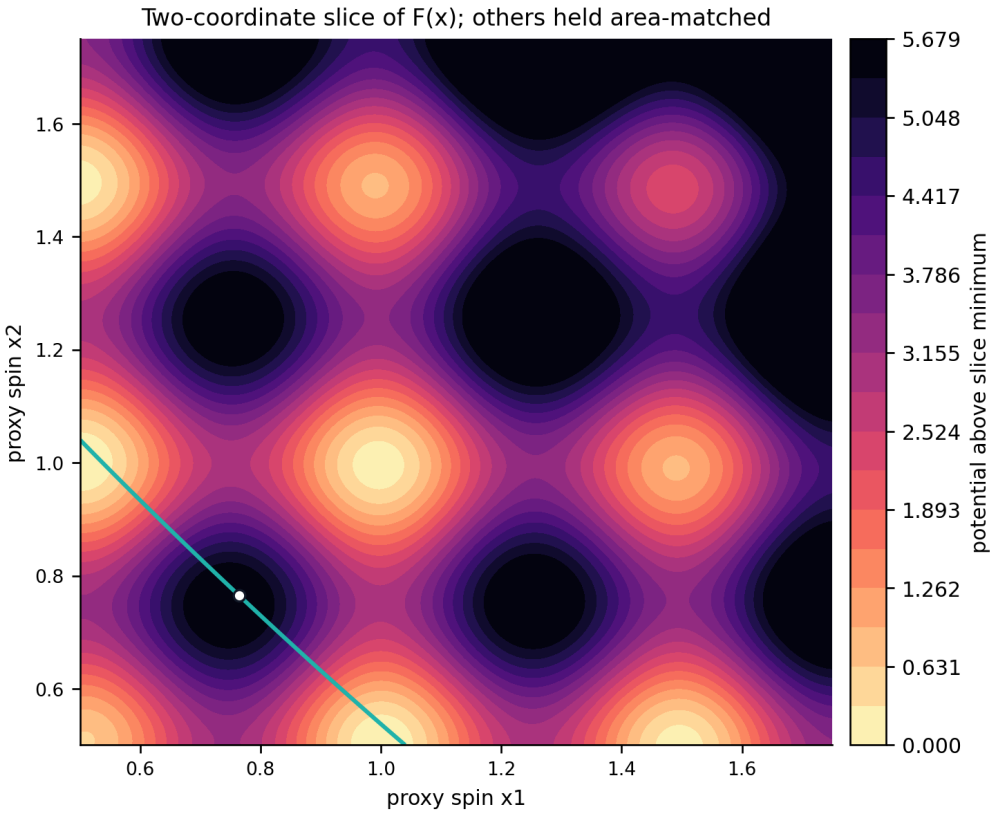
$$H_x = \kappa_A\, C^2\, vv^T + \mathrm{diag}(d_i) \qquad (17)$$

$$d_i = \kappa_A\, C\, r\, a''(x_i) + 8\pi^2 \kappa_Q \cos(4\pi x_i) + 4\eta/(2x_i+1)^2 \qquad (18)$$

where a''(x)=-1/[4{x(x+1)}^{3/2}]. The rank-one term is the stiff normal direction to the area shell; most remaining directions are tangential rearrangements of puncture spins. Negative comb curvature near barrier tops creates saddle regions even though the global target is proper. The latent Hessian additionally contains chain-rule terms from the Softplus map and Jacobian.

Figure 4

Two-Coordinate Slice of the Relaxed Physical Potential



Note. This slice visualizes geometry; it is not a marginal posterior density because the other coordinates are fixed rather than integrated out.

Ergodicity, Metastability, and Sector Mixing

Irreducibility Versus Practical Mixing

The corrected density is positive and smooth on connected R^N. Gaussian momentum refreshment has full support, and randomized finite trajectories avoid a deterministic resonance schedule. Under standard regularity and nondegeneracy conditions, these properties support irreducibility and aperiodicity of the Metropolis-corrected chain. They do not bound the spectral gap uniformly in N or kappa_Q. A formally ergodic chain may remain in one collection of lattice wells for the entire feasible run.

Coordinated Barriers

Changing one spin by one-half generally perturbs total area. Remaining near the area shell can require an opposing change in another coordinate or a coordinated path across several comb barriers. If m coordinates must cross near their maxima, quantization energy alone can rise by approximately m kappa_Q. The probability of a momentum refreshment supplying that energy declines rapidly, and the path must also find a dynamically aligned direction. Local acceptance can therefore be excellent while global spin-sector mixing is poor.

Replica Strategy

A confirmatory sampler should temper only the comb while keeping the macroscopic shell identifiable:

$$\pi_r(x) \propto \exp\{-U_macro(x) - U_deg(x) - \tau_r \, \kappa_Q \, Q(x)\} \qquad (19)$$

with 0=tau_0<...<tau_R=1. Low-tau replicas cross spin sectors; adjacent swaps transport those sectors to the target. Swap acceptance, round-trip time, sector-transition counts, and replica-specific ESS must be reported. Simulated tempering or annealed importance sampling are alternatives when the normalizing-constant ratios are estimated consistently (Neal, 2001).

High-Dimensional Scaling and Preconditioning

For regular product targets, asymptotic HMC theory gives epsilon proportional to N^{-1/4} to maintain an order-one acceptance probability and an asymptotically efficient acceptance near .651 (Beskos et al., 2013). The relaxed horizon is not a product target: its area shell induces global dependence and the comb induces multimodality. Nevertheless, N^{-1/4} is a useful stress-test reference. Every scaling study should report ESS per gradient evaluation, memory, wall time, energy error, and spin-sector transitions rather than raw acceptance alone.

Diagonal Plus Rank-One Metric

A diagonal mass adapts marginal scales but does not directly align with the area-normal direction. The Hessian structure motivates

$$M = D + \rho u u^{T}, \qquad u \propto \nabla_z A \qquad (20)$$

$$M^{-1} = D^{-1} - [\rho D^{-1} u u^{T} D^{-1}]/[1 + \rho u^{T} D^{-1} u] \qquad (21)$$

so velocity and kinetic energy can be evaluated in O(N) time by Sherman-Morrison. Adaptation must be regularized and frozen after warmup. A block metric grouping punctures by provisional spin sector is another candidate, but sector labels derived from rounded states may change discontinuously. Riemannian HMC could track local curvature more closely, at substantially greater computational cost (Girolami & Calderhead, 2011).

NUTS Compatibility

The potential and exact gradients can be supplied to Pyro or another PyTorch-native NUTS backend. NUTS removes manual trajectory length but does not eliminate multimodal barriers. Tree depth, divergent transitions, energy Bayesian fraction of missing information, and rank-normalized diagnostics should accompany NUTS output.

Computational Methods and Baseline Design

The repository implements the model in pure PyTorch with float64 state tensors and explicit reverse-mode autograd. A deterministic seed initializes a nearly homogeneous area-matched state. Warmup applies dual averaging and Welford marginal-variance estimates; retained transitions use fixed tuning. Atomic JSON and tensor artifacts record configuration hashes, runtime versions, and diagnostic summaries. Unit tests cover the lower bound, comb zeros, tail propriety, analytic/autograd gradient agreement, initialization, rounding, artifact round trips, leapfrog involution, and finite short-chain output. Continuous integration passes under Python 3.11 and 3.12.

Table 2

Validated Baseline Configuration

Quantity	Symbol	Baseline
Punctures	N	64
Target area	A_target	512.0
Barbero-Immirzi parameter	gamma	0.274
Area penalty	kappa_A	0.05
Comb strength	kappa_Q	3.0
Degeneracy exponent	eta	0.5
Warmup / retained	W / S	80 / 80
Initial leapfrog step	epsilon_0	0.004
Leapfrog steps	L	5 to 12, randomized
Target acceptance	delta	0.85
Seed	-	20260823

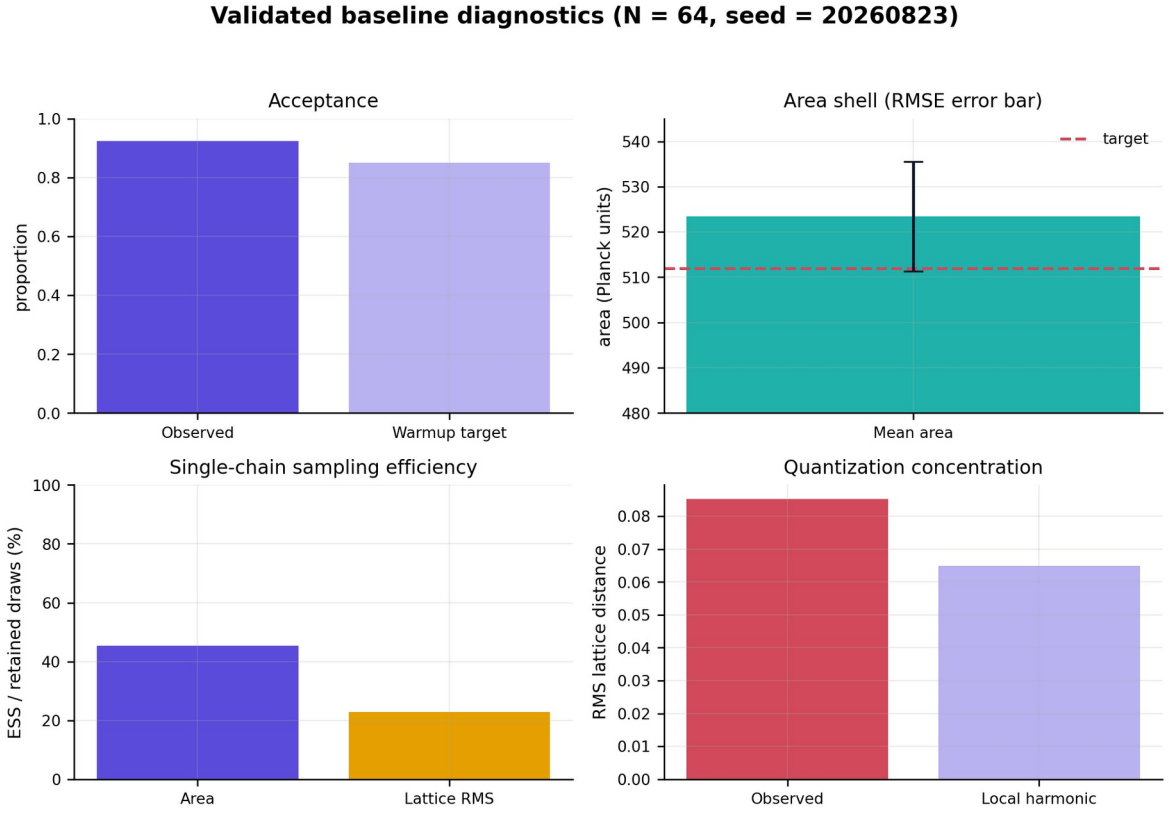
Note. The run is a numerical feasibility study, not a thermodynamic-limit estimate. Only 80 retained transitions from one chain were used.

Validated Baseline Findings

The N=64 chain accepted 74 of 80 retained proposals (acceptance .925), had mean acceptance probability .8865, and recorded no divergences. Mean absolute Hamiltonian error was .2230 and the maximum was 1.2186. Warmup expanded the initial step size from .004 to .07860, showing that the original step was deliberately conservative. These diagnostics establish stable local integration for this configuration.

Figure 5

Validated HMC, Area-Shell, and Lattice Diagnostics



The zero-divergence result should not be generalized beyond the sampled region. Divergences diagnose numerical integration failure; they do not reveal a chain that never attempted to cross a distant comb barrier.

Area Control, Quantization, and Sampling Efficiency

Table 3

Baseline Numerical Findings

Diagnostic	Estimate	Interpretation
Acceptance rate	0.925	Stable but conservative trajectories
Divergences	0 / 80	No detected numerical singularity
Mean |delta H|	0.2230	Finite integration error
Maximum |delta H|	1.2186	No divergence under threshold
Final step size	0.07860	Warmup expanded initial step
Mean area	523.405	2.23% above target
Area RMSE	12.141	Soft shell remains visibly broad
Area ESS	36.22 / 80	45.3% efficiency
Mean lattice RMS	0.08525	Near, but not on, the lattice
Lattice ESS	18.34 / 80	22.9% efficiency; slow mode

Note. ESS values use the repository's single-chain initial-positive autocorrelation estimator and are exploratory.

Mean area exceeded the target by 11.4053, or 2.23%, and area RMSE was 12.1412. The soft area penalty therefore kept the chain near the shell without reproducing a sharply microcanonical ensemble. Area ESS was 36.22 of 80 draws (45.3%). The mean root-mean-square distance to the nearest half-integer vector was .08525, but its ESS was only 18.34 (22.9%), identifying quantization structure as the slower monitored mode.

The local harmonic prediction at kappa_Q=3 is .06497; the observed .08525 is about 31% larger. This discrepancy is expected when area and degeneracy forces shift local modes and when draws span nonquadratic portions of the wells. It is informative evidence about the finite relaxation, not a failure of the Taylor expansion.

What the Baseline Establishes

The project demonstrates a proper differentiable target, accurate gradients, stable HMC integration, and measurable lattice concentration at moderate dimension. It does not establish global equilibrium, exact discrete probabilities, the thermodynamic limit, or black-hole entropy. Multiple dispersed chains and sector-aware diagnostics are necessary before the posterior itself can be treated as numerically resolved.

From Relaxed Samples to a Discrete Normalizer

Let F_0 contain the area and degeneracy terms and Q(x)=sum_i sin^2(2 pi x_i). The relaxed normalizer is

$$Z(\kappa_Q) = \int \exp[-F_0(x) - \kappa_Q\, Q(x)]\, dx \qquad (22)$$

$$d \log Z(\kappa_Q)/d\kappa_Q = -E_\{\kappa_Q\}[Q(X)] \qquad (23)$$

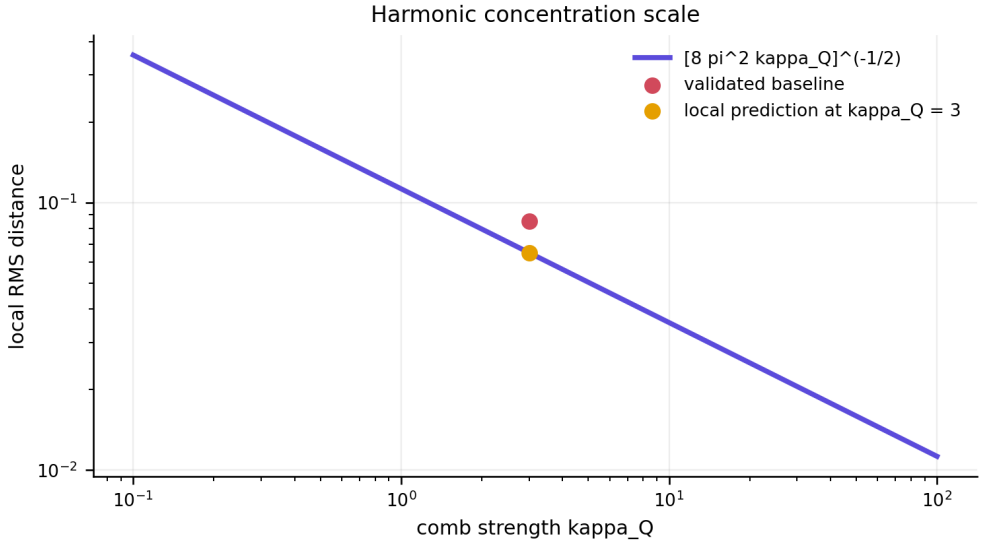
Thermodynamic integration estimates log-normalizer differences by quadrature of the expected comb statistic. An absolute result still requires a calibrated reference Z(0). In the strong-comb limit, each interior well contributes a Gaussian volume approximately [2 sqrt(pi kappa_Q)]^{-N}. Thus a discrete lattice sum is related to a rescaled, not raw, continuous normalizer:

$$\sum_h \exp[-F_0(h)] \approx [4\pi\kappa_Q]\hat{}(N/2)\, Z(\kappa_Q) \qquad (24)$$

with boundary, overlap, and finite-kappa_Q corrections. Exact physical counting then replaces the proxy degeneracy by d_phys(h). Small-N systems that can be enumerated exactly are indispensable for calibrating this pipeline and measuring bias in rounded or basin-integrated estimates (Gelman & Meng, 1998).

Figure 6

Predicted Local Concentration Across Comb Strength



Validity Analysis and Identifiability Limits

The relaxed model contains several parameters that can compensate for one another. Larger kappa_A tightens the area shell but raises collective curvature; larger kappa_Q improves local lattice concentration but suppresses sector transitions; larger eta favors higher spins and can move area burden among fewer punctures. With N fixed, these effects can be partly separated through replicated design. If N is also unknown, puncture count and spin magnitude become structurally confounded under an area-only observation. Additional observables or explicit priors are required.

Table 4

Principal Threats to Physical and Statistical Validity

Threat	Consequence	Required remedy
Finite comb barriers	Metastable spin sectors	Replica exchange over comb strength
Single short chain	No reliable convergence claim	At least four dispersed chains and rank R-hat
Proxy degeneracy	Biased physical state weights	Exact intertwiner/Chern-Simons correction
Soft area penalty	Non-microcanonical area spread	Window sensitivity or constrained dynamics
Unknown normalizer	No absolute entropy	Thermodynamic integration with calibrated reference
Fixed N	Missed puncture-number uncertainty	Grand-canonical or reversible-jump extension

Note. A successful HMC diagnostic does not repair a misspecified physical multiplicity or an unidentified ensemble.

Single-chain ESS is especially fragile for multimodal targets. Rank-normalized split R-hat, bulk and tail ESS, Monte Carlo standard errors, and rank plots require multiple chains (Vehtari et al., 2021). Posterior predictive checks should include area distributions, spin occupancy, coordinated transition counts, and exact small-N frequencies. Simulation-based calibration can then test whether a Bayesian extension recovers known generating parameters across replicated synthetic ensembles.

Confirmatory Research Program and Conclusions

Table 5

Proposed Scaling and Falsification Design

Axis	Planned levels	Primary diagnostic
Dimension N	64, 128, 256, 512, 1024, 2048	ESS per gradient and memory
Comb kappa_Q	0, 0.5, 1, 2, 3, 5, 8	Lattice RMS and sector transitions
Area kappa_A	0.01, 0.05, 0.2	Area RMSE and stiffness
Metric	Identity, diagonal, diagonal + rank one	Divergences and efficiency
Degeneracy	Proxy, surrogate, exact small-N	Bias in discrete probabilities
Validation	Exactly enumerable N	Total variation and normalizer error

Decision Criteria

The relaxation will be considered computationally successful only if: (a) at least four dispersed target chains have rank-normalized R-hat below 1.01; (b) bulk and tail ESS per gradient remain usable as N grows; (c) tempered chains complete repeated round trips; (d) small-N corrected probabilities agree with enumeration within predeclared total-variation and normalizer tolerances; and (e) conclusions are stable across area-window, comb, and metric choices. Failure on any criterion falsifies the claim that the current relaxation provides a reliable route to high-dimensional microstate weights.

Conclusion

The project resolves the first mathematical obstacle to differentiable horizon sampling: it defines a proper latent measure and supplies exact gradients for an LQG-inspired area ensemble. The finite comb preserves smooth Hamiltonian paths while assigning positive probability to barrier crossing, and the rank-one area geometry suggests scalable preconditioning. Baseline results show stable local computation and nontrivial half-integer concentration. The same results also expose the principal limitation: lattice-distance mixing is substantially slower than area mixing. The defensible conclusion is therefore feasibility, not entropy. Physical inference requires tempered global exploration, normalizing-constant estimation, and exact invariant-state correction.

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Appendix A

Python Code for All Figures and Tables

The following script is reproduced verbatim. It generates Figures 1-6 and Tables 1-5. Baseline values are the validated seed-20260823 repository summary; analytical figures introduce no stochastic data.

```
"""Generate every numerical figure and table used in the Problem 2 report.

The figures are deterministic analytical evaluations of the documented model,
except Figure 5 and Table 3, which reproduce the validated baseline summary
from seed 20260823.  No unreported stochastic simulation is introduced here.
"""

from __future__ import annotations

import csv
import math
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.patches import FancyArrowPatch, FancyBboxPatch


ROOT = Path(__file__).resolve().parents[1]
FIGURES = ROOT / "figures"
TABLES = ROOT / "tables"
FIGURES.mkdir(parents=True, exist_ok=True)
TABLES.mkdir(parents=True, exist_ok=True)

GAMMA = 0.274
LP2 = 1.0
N = 64
AREA_TARGET = 512.0
KAPPA_AREA = 0.05
KAPPA_QUANT = 3.0
ETA = 0.5
C = 8.0 * math.pi * GAMMA * LP2

BASELINE = {
    "retained_draws": 80,
    "warmup_steps": 80,
    "acceptance_rate": 0.925,
    "mean_acceptance_probability": 0.886455,
    "divergences": 0,
    "mean_abs_energy_error": 0.222999,
    "max_abs_energy_error": 1.218586,
    "final_step_size": 0.0786007,
    "area_mean": 523.4053,
    "area_rmse": 12.1412,
    "area_ess": 36.2164,
    "lattice_rms": 0.0852524,
    "lattice_ess": 18.3371,
}


def style_axes(ax, *, grid: bool = True) -> None:
    ax.spines[["top", "right"]].set_visible(False)
    if grid:
        ax.grid(alpha=0.22, linewidth=0.7)
    ax.tick_params(labelsize=8)


def save(fig, name: str) -> None:
    fig.savefig(FIGURES / name, dpi=240, bbox_inches="tight", facecolor="white")
    plt.close(fig)


def write_csv(name: str, rows: list[list[str]]) -> None:
    with (TABLES / name).open("w", newline="", encoding="utf-8") as stream:
        csv.writer(stream).writerows(rows)


plt.rcParams.update(
    {
        "font.family": "DejaVu Sans",
        "font.size": 9,
        "axes.titlesize": 10,
        "axes.labelsize": 9,
        "figure.titlesize": 12,
    }
)
```

Appendix A

Python Code for All Figures and Tables

Appendix A continued (2 of 4).

```
# Figure 1: computational/statistical architecture.
fig, ax = plt.subplots(figsize=(9.2, 4.6))
ax.set_xlim(0, 10)
ax.set_ylim(0, 6)
ax.axis("off")
nodes = [
    (0.4, 2.25, 1.65, 1.25, "Latent z", "unconstrained"),
    (2.45, 2.25, 1.75, 1.25, "Proxy spins x", "Softplus map"),
    (4.65, 3.55, 2.0, 1.35, "Physical target", "area + comb + proxy"),
    (4.65, 0.95, 2.0, 1.35, "Measure correction", "Softplus Jacobian"),
    (7.05, 2.25, 1.65, 1.25, "Potential U", "proper latent density"),
    (8.95, 2.25, 0.85, 1.25, "HMC", "draws"),
]
for x0, y0, w, h, title, subtitle in nodes:
    patch = FancyBboxPatch(
        (x0, y0), w, h, boxstyle="round,pad=0.03,rounding_size=0.12",
        facecolor="#F5F3FF", edgecolor="#5B4BDB", linewidth=1.3
    )
    ax.add_patch(patch)
    ax.text(x0 + w / 2, y0 + 0.73 * h, title, ha="center", va="center", weight="bold")
    ax.text(x0 + w / 2, y0 + 0.34 * h, subtitle, ha="center", va="center", fontsize=7.8)
for start, end in [
    ((2.05, 2.88), (2.45, 2.88)),
    ((4.2, 2.88), (4.65, 4.22)),
    ((4.2, 2.88), (4.65, 1.62)),
    ((6.65, 4.22), (7.05, 3.1)),
    ((6.65, 1.62), (7.05, 2.65)),
    ((8.7, 2.88), (8.95, 2.88)),
]:
    ax.add_patch(FancyArrowPatch(start, end, arrowstyle="-|>", mutation_scale=12,
                                 color="#374151", linewidth=1.2))
ax.text(5.65, 5.35, "F(x)", ha="center", weight="bold", color="#5B4BDB")
ax.text(5.65, 0.55, "-sum log sigmoid(z)", ha="center", weight="bold", color="#5B4BDB")
ax.set_title("Differentiable quantum-horizon ensemble", pad=10, weight="bold")
save(fig, "figure_1_architecture.png")

# Figure 2: transform and Jacobian.
z = np.linspace(-8, 6, 600)
softplus = np.logaddexp(0.0, z)
x = 0.5 + softplus
sigmoid = 1.0 / (1.0 + np.exp(-z))
neg_log_jac = -np.log(sigmoid)
fig, axes = plt.subplots(1, 2, figsize=(9.0, 3.6))
axes[0].plot(z, x, color="#5B4BDB", linewidth=2.2)
axes[0].axhline(0.5, color="#E59F00", linestyle="--", label="lower bound 1/2")
axes[0].set(xlabel="latent coordinate z", ylabel="proxy spin x(z)", title="Softplus proxy map")
axes[0].legend(frameon=False, fontsize=8)
style_axes(axes[0])
axes[1].plot(z, neg_log_jac, color="#D1495B", linewidth=2.2)
axes[1].set(xlabel="latent coordinate z", ylabel="-log sigmoid(z)",
            title="Jacobian confinement of the negative tail")
style_axes(axes[1])
fig.tight_layout()
save(fig, "figure_2_softplus_jacobian.png")

# Figure 3: periodic comb and its harmonic approximation.
xg = np.linspace(0.5, 3.0, 1000)
comb = KAPPA_QUANT * np.sin(2.0 * math.pi * xg) ** 2
h = 1.5
delta = np.linspace(-0.22, 0.22, 500)
exact_local = KAPPA_QUANT * np.sin(2.0 * math.pi * (h + delta)) ** 2
quadratic = 4.0 * math.pi**2 * KAPPA_QUANT * delta**2
fig, axes = plt.subplots(1, 2, figsize=(9.0, 3.6))
axes[0].plot(xg, comb, color="#5B4BDB", linewidth=2)
for q in np.arange(0.5, 3.01, 0.5):
    axes[0].axvline(q, color="#9CA3AF", linewidth=0.45, alpha=0.55)
axes[0].set(xlabel="proxy spin x", ylabel="quantization energy",
            title="Finite half-integer comb wells")
style_axes(axes[0])
axes[1].plot(delta, exact_local, label="exact comb", color="#5B4BDB", linewidth=2)
axes[1].plot(delta, quadratic, label="quadratic approximation", color="#E59F00",
             linestyle="--", linewidth=2)
axes[1].set(xlabel="delta from h = 3/2", ylabel="quantization energy",
            title="Local well geometry")
axes[1].legend(frameon=False, fontsize=8)
style_axes(axes[1])
fig.tight_layout()
```

Appendix A

Python Code for All Figures and Tables

Appendix A continued (3 of 4).

```
save(fig, "figure_3_comb_geometry.png")

# Figure 4: two-coordinate physical potential slice, holding N-2 coordinates fixed.
area_per = AREA_TARGET / (C * N)
x_bar = 0.5 * (-1.0 + math.sqrt(1.0 + 4.0 * area_per**2))
grid = np.linspace(0.5005, 1.75, 360)
x1, x2 = np.meshgrid(grid, grid)
rest_area = (N - 2) * math.sqrt(x_bar * (x_bar + 1.0))
area = C * (np.sqrt(x1 * (x1 + 1.0)) + np.sqrt(x2 * (x2 + 1.0)) + rest_area)
macro = 0.5 * KAPPA_AREA * (area - AREA_TARGET) ** 2
quant = KAPPA_QUANT * (np.sin(2.0 * math.pi * x1) ** 2 + np.sin(2.0 * math.pi * x2) ** 2)
deg = -ETA * (np.log(2.0 * x1 + 1.0) + np.log(2.0 * x2 + 1.0))
potential = macro + quant + deg
potential -= np.nanmin(potential)
fig, ax = plt.subplots(figsize=(6.3, 5.1))
levels = np.linspace(0, min(18.0, np.percentile(potential, 85)), 19)
cont = ax.contourf(x1, x2, np.clip(potential, 0, levels[-1]), levels=levels, cmap="magma_r")
ax.contour(x1, x2, area, levels=[AREA_TARGET], colors="#20B2AA", linewidths=1.8)
ax.plot([x_bar], [x_bar], marker="o", color="white", markeredgecolor="#111827", markersize=5)
ax.set(xlabel="proxy spin x1", ylabel="proxy spin x2",
       title="Two-coordinate slice of F(x); others held area-matched")
style_axes(ax, grid=False)
cbar = fig.colorbar(cont, ax=ax, pad=0.02)
cbar.set_label("potential above slice minimum")
fig.tight_layout()
save(fig, "figure_4_potential_slice.png")

# Figure 5: validated baseline metrics.
harmonic_rms = 1.0 / math.sqrt(8.0 * math.pi**2 * KAPPA_QUANT)
fig, axes = plt.subplots(2, 2, figsize=(8.7, 6.1))
axes[0, 0].bar(["Observed", "Warmup target"], [BASELINE["acceptance_rate"], 0.85],
               color=["#5B4BDB", "#B8B3F0"])
axes[0, 0].set_ylim(0, 1.0)
axes[0, 0].set_ylabel("proportion")
axes[0, 0].set_title("Acceptance")
style_axes(axes[0, 0])
axes[0, 1].bar(["Mean area"], [BASELINE["area_mean"]], color="#20B2AA")
axes[0, 1].axhline(AREA_TARGET, color="#D1495B", linestyle="--", label="target")
axes[0, 1].errorbar([0], [BASELINE["area_mean"]], yerr=[BASELINE["area_rmse"]],
                    fmt="none", ecolor="#111827", capsize=5)
axes[0, 1].set_ylim(480, 545)
axes[0, 1].set_ylabel("area (Planck units)")
axes[0, 1].set_title("Area shell (RMSE error bar)")
axes[0, 1].legend(frameon=False, fontsize=8)
style_axes(axes[0, 1])
ess_pct = [100 * BASELINE["area_ess"] / 80, 100 * BASELINE["lattice_ess"] / 80]
axes[1, 0].bar(["Area", "Lattice RMS"], ess_pct, color=["#5B4BDB", "#E59F00"])
axes[1, 0].set_ylim(0, 100)
axes[1, 0].set_ylabel("ESS / retained draws (%)")
axes[1, 0].set_title("Single-chain sampling efficiency")
style_axes(axes[1, 0])
axes[1, 1].bar(["Observed", "Local harmonic"], [BASELINE["lattice_rms"], harmonic_rms],
               color=["#D1495B", "#B8B3F0"])
axes[1, 1].set_ylabel("RMS lattice distance")
axes[1, 1].set_title("Quantization concentration")
style_axes(axes[1, 1])
fig.suptitle("Validated baseline diagnostics (N = 64, seed = 20260823)", weight="bold")
fig.tight_layout(rect=[0, 0, 1, 0.95])
save(fig, "figure_5_baseline.png")

# Figure 6: predicted local well scale across comb strengths.
kappa = np.logspace(-1, 2, 300)
well_sd = 1.0 / np.sqrt(8.0 * math.pi**2 * kappa)
fig, ax = plt.subplots(figsize=(6.4, 3.7))
ax.loglog(kappa, well_sd, color="#5B4BDB", linewidth=2.3,
          label="[8 pi^2 kappa_Q]^(-1/2)")
ax.scatter([KAPPA_QUANT], [BASELINE["lattice_rms"]], color="#D1495B", s=42,
           zorder=3, label="validated baseline")
ax.scatter([KAPPA_QUANT], [harmonic_rms], color="#E59F00", s=42,
           zorder=3, label="local prediction at kappa_Q = 3")
ax.set(xlabel="comb strength kappa_Q", ylabel="local RMS distance",
       title="Harmonic concentration scale")
ax.legend(frameon=False, fontsize=8)
style_axes(ax)
fig.tight_layout()
save(fig, "figure_6_concentration_scaling.png")
```

Appendix A

Python Code for All Figures and Tables

Appendix A continued (4 of 4).

```
write_csv(
    "table_1_model_components.csv",
    [
        ["Component", "Mathematical role", "Interpretive status"],
        ["Discrete spins j_i", "Half-integer horizon labels", "Exact LQG kinematics"],
        ["Proxy spins x_i", "Smooth positive relaxation", "Computational surrogate"],
        ["Area penalty", "Soft target-area shell", "Not a hard microcanonical constraint"],
        ["Periodic comb", "Concentrates near half-integers", "Finite-barrier regularizer"],
        ["Product (2x_i+1)^eta", "Local degeneracy pressure", "Not SU(2) intertwiner dimension"],
        ["Softplus Jacobian", "Defines correct latent measure", "Required for propriety"],
    ],
)

write_csv(
    "table_2_configuration.csv",
    [
        ["Quantity", "Symbol", "Baseline"],
        ["Punctures", "N", "64"],
        ["Target area", "A_target", "512.0"],
        ["Barbero-Immirzi parameter", "gamma", "0.274"],
        ["Area penalty", "kappa_A", "0.05"],
        ["Comb strength", "kappa_Q", "3.0"],
        ["Degeneracy exponent", "eta", "0.5"],
        ["Warmup / retained", "W / S", "80 / 80"],
        ["Initial leapfrog step", "epsilon_0", "0.004"],
        ["Leapfrog steps", "L", "5 to 12, randomized"],
        ["Target acceptance", "delta", "0.85"],
        ["Seed", "-", "20260823"],
    ],
)

write_csv(
    "table_3_baseline_findings.csv",
    [
        ["Diagnostic", "Estimate", "Interpretation"],
        ["Acceptance rate", "0.925", "Stable but conservative trajectories"],
        ["Divergences", "0 / 80", "No detected numerical singularity"],
        ["Mean |delta H|", "0.2230", "Finite integration error"],
        ["Maximum |delta H|", "1.2186", "No divergence under threshold"],
        ["Final step size", "0.07860", "Warmup expanded initial step"],
        ["Mean area", "523.405", "2.23% above target"],
        ["Area RMSE", "12.141", "Soft shell remains visibly broad"],
        ["Area ESS", "36.22 / 80", "45.3% efficiency"],
        ["Mean lattice RMS", "0.08525", "Near, but not on, the lattice"],
        ["Lattice ESS", "18.34 / 80", "22.9% efficiency; slow mode"],
    ],
)

write_csv(
    "table_4_validity.csv",
    [
        ["Threat", "Consequence", "Required remedy"],
        ["Finite comb barriers", "Metastable spin sectors", "Replica exchange over comb strength"],
        ["Single short chain", "No reliable convergence claim", "At least four dispersed chains and rank R-hat"],
        ["Proxy degeneracy", "Biased physical state weights", "Exact intertwiner/Chern-Simons correction"],
        ["Soft area penalty", "Non-microcanonical area spread", "Window sensitivity or constrained dynamics"],
        ["Unknown normalizer", "No absolute entropy", "Thermodynamic integration with calibrated reference"],
        ["Fixed N", "Missed puncture-number uncertainty", "Grand-canonical or reversible-jump extension"],
    ],
)

write_csv(
    "table_5_confirmatory_design.csv",
    [
        ["Axis", "Planned levels", "Primary diagnostic"],
        ["Dimension N", "64, 128, 256, 512, 1024, 2048", "ESS per gradient and memory"],
        ["Comb kappa_Q", "0, 0.5, 1, 2, 3, 5, 8", "Lattice RMS and sector transitions"],
        ["Area kappa_A", "0.01, 0.05, 0.2", "Area RMSE and stiffness"],
        ["Metric", "Identity, diagonal, diagonal + rank one", "Divergences and efficiency"],
        ["Degeneracy", "Proxy, surrogate, exact small-N", "Bias in discrete probabilities"],
        ["Validation", "Exactly enumerable N", "Total variation and normalizer error"],
    ],
)

print(f"Wrote {len(list(FIGURES.glob('*.png')))} figures and {len(list(TABLES.glob('*.csv')))} tables")
```